1. What is Artificial Neural Network (ANN)?

Answer:

> Artificial Neural Network is a computational model inspired by the structure and working of the human brain. It consists of interconnected neurons that process information and learn patterns from data using weights and activation functions.

Deeper understanding:
ANN takes input data, processes it through layers of neurons, and produces output. During training, the network adjusts weights and bias to minimize error and improve prediction accuracy.

Applications:

- Image recognition
- Speech recognition
- Recommendation systems
- Medical diagnosis

2. Why is ANN called brain-inspired computing?

Answer:

> ANN is called brain-inspired computing because its structure is loosely based on biological neurons present in the human brain.

Deeper understanding:
In the human brain, neurons communicate through electrical signals. Similarly, artificial neurons receive input, process it mathematically, and pass output to other neurons.

Difference:

- Human brain neurons are biological
- ANN neurons are mathematical models implemented in software

3. What are neurons in ANN?

Answer:

> A neuron is the basic computational unit of a neural network. It receives multiple inputs, multiplies them with weights, adds bias, applies activation function, and generates output.

Neuron equation:

y=f(∑wx+b)y=f(\sum wx+b)y=f(∑wx+b)

Where:

- xxx = input
- www = weights
- bbb = bias
- fff = activation function

Deeper understanding:
Each neuron learns features from data. Lower layers learn simple patterns while deeper layers learn complex patterns.

4. What is a perceptron?

Answer:

> Perceptron is the simplest single-layer neural network used for binary classification problems.

Deeper understanding:
Perceptron takes inputs, calculates weighted sum, applies activation function, and predicts output.

Decision rule:

y={1,wx+b≥00,wx+b<0y=\begin{cases}1,&wx+b\ge0\\0,&wx+b<0\end{cases}y={1,0,wx+b≥0wx+b<0

Important point:
Perceptron can solve only linearly separable problems.

Examples:

- Spam or not spam
- Pass or fail
- Yes or no classification

5. Difference between biological neuron and artificial neuron?

Biological Neuron	Artificial Neuron
Exists in human brain	Exists in computer model
Uses electrochemical signals	Uses mathematical operations
Learns naturally	Learns using algorithms
Very complex	Simplified model

Deeper understanding:
Artificial neurons are inspired by biological neurons but are much simpler mathematically.

6. What are weights and bias?

Answer:

> Weights determine the importance of input features, while bias helps shift the activation function output.

Deeper understanding:

- Larger weight → more influence of that input
- Smaller weight → less influence

Bias helps the model fit data more flexibly.

Equation:

z=w1x1+w2x2+bz=w_1x_1+w_2x_2+bz=w1x1+w2x2+b

Without bias, the decision boundary always passes through origin, reducing flexibility.

7. What is activation function?

Answer:

> Activation function determines whether a neuron should activate and introduces non-linearity into the neural network.

Deeper understanding:
Without activation function, even deep networks behave like linear models and cannot solve complex problems.

Common activation functions:

- Sigmoid
- ReLU
- Tanh
- Softmax

Most common:
ReLU

f(x)=max(0,x)f(x)=\max(0,x)f(x)=max(0,x)

8. Why activation function is needed?

Answer:

> Activation function is needed to introduce non-linearity so that neural networks can learn complex relationships in data.

Deeper understanding:
Real-world data is mostly non-linear. Without activation functions:

- neural networks cannot recognize images properly
- cannot solve XOR problem
- cannot learn complex patterns

Activation functions help deep learning models become powerful.

9. Types of activation functions?

Sigmoid

σ(x)=11+e−x\sigma(x)=\frac{1}{1+e^{-x}}σ(x)=1+e−x1

- Output between 0 and 1
- Used in binary classification

ReLU

f(x)=max(0,x)f(x)=\max(0,x)f(x)=max(0,x)

- Most widely used
- Fast and efficient

Tanh

tanh(x)=ex−e−xex+e−xtanh(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}}tanh(x)=ex+e−xex−e−x

- Output between -1 and 1

Softmax

Used in multi-class classification.

10. What is learning in ANN?

Answer:

> Learning in ANN means adjusting weights and bias to minimize error and improve prediction accuracy.

Deeper understanding:
The network:

1. predicts output
2. compares with actual output
3. calculates error
4. updates weights
5. repeats until error becomes minimum

Perceptron learning rule:

wnew=wold+η(y−y^)xw_{new}=w_{old}+\eta(y-\hat{y})xwnew=wold+η(y−y^)x

Where:

- η\etaη = learning rate
- yyy = actual output
- y^\hat{y}y^ = predicted output

This process is repeated over multiple epochs until the model converges.

11. What is perceptron learning law?

Answer:

> Perceptron learning law is a rule used to update weights and bias whenever the perceptron makes a wrong prediction.

Learning rule:

wnew=wold+η(y−y^)xw_{new}=w_{old}+\eta(y-\hat{y})xwnew=wold+η(y−y^)x

Where:

- η\etaη = learning rate
- yyy = actual output
- y^\hat{y}y^ = predicted output

Deeper understanding:
If prediction is correct, weights remain same.
If prediction is wrong, weights are adjusted to reduce future error.

Purpose:

- Improve classification accuracy
- Move decision boundary toward correct classification

---

12. What is the goal of perceptron?

Answer:

> The main goal of perceptron is to classify input data into different classes using a decision boundary.

Deeper understanding:
Perceptron tries to separate data points into categories such as:

- 0 or 1
- spam or not spam
- pass or fail

It learns a linear boundary that separates classes correctly.

---

13. Why perceptron is called linear classifier?

Answer:

Perceptron is called a linear classifier because it separates data using a straight line or linear decision boundary.

Decision boundary equation:

w1x1+w2x2+b=0w_1x_1+w_2x_2+b=0w1x1+w2x2+b=0

Deeper understanding:
In 2D:

- boundary is a straight line

In 3D:

- boundary is a plane

Perceptron cannot classify non-linear data like XOR.

---

14. What is decision boundary?

Answer:

Decision boundary is the line, plane, or surface that separates different classes in a classification problem.

Deeper understanding:
The perceptron creates a boundary between classes.

Example:

- Blue points on one side
- Red points on another side

In your graph:

- dashed line = decision boundary
- colored regions = decision regions

---

15. What are decision regions?

Answer:

Decision regions are areas in feature space where the model predicts a particular class.

Deeper understanding:
Each side of the decision boundary belongs to one class.

Example:

- Blue region → Class 0
- Red region → Class 1

Decision regions help visualize how the classifier works.

---

16. Why perceptron cannot solve XOR problem?

Answer:

Perceptron cannot solve XOR because XOR data is not linearly separable.

Deeper understanding:
No single straight line can separate XOR outputs correctly.

XOR truth table:

Input	Output
0 0	0
0 1	1
1 0	1
1 1	0

This requires non-linear separation.

Solution:

- Multi-layer perceptron (MLP)
- Hidden layers
- Non-linear activation functions

---

17. What happens when prediction is wrong?

Answer:

When prediction is wrong, the perceptron updates weights and bias using the learning rule.

Deeper understanding:
Wrong predictions produce error.

Then:

- weights are adjusted
- bias is updated
- decision boundary shifts

This improves future predictions.

---

18. What is learning rate?

Answer:

Learning rate controls how much weights are updated during training.

Symbol:

η\etaη

Deeper understanding:

- Small learning rate → slow learning
- Large learning rate → unstable learning

If learning rate is too high:

- model may overshoot optimal solution

If too low:

- training becomes very slow

---

19. What is epoch?

Answer:

One complete pass of the entire training dataset through the neural network is called an epoch.

Deeper understanding:
Suppose dataset has 1000 samples.

If all 1000 samples are processed once:

- 1 epoch completed

Training usually requires multiple epochs.

Purpose:

- gradually reduce error
- improve accuracy

---

20. What is convergence?

Answer:

> Convergence means the model has learned successfully and errors stop reducing significantly.

Deeper understanding:
In perceptron:

- convergence occurs when misclassification becomes zero

Example:

Epoch 4: misclassifications = 0

Meaning:

- model found correct decision boundary
- training can stop

Important point:
Perceptron converges only if data is linearly separable

21. Explain sigmoid activation function.

Answer:

> Sigmoid is an activation function that converts input values into a range between 0 and 1.

Equation:

$$\sigma(x)=\frac{1}{1+e^{-x}}$$

Deeper understanding:

- Large positive input → output near 1
- Large negative input → output near 0

Uses:

- Binary classification
- Logistic regression

Limitation:

- Suffers from vanishing gradient problem

---

22. Difference between sigmoid and ReLU?

Sigmoid	**ReLU**
Output between 0 and 1	Output is 0 or positive value
Slower training	Faster training
Vanishing gradient problem	Less vanishing gradient
Used in output layer	Mostly used in hidden layers

ReLU equation:

$$f(x)=\max(0,x)$$

Deeper understanding:
ReLU is preferred in deep learning because it trains networks faster.

---

23. Why ReLU is mostly used?

Answer:

ReLU is mostly used because it is simple, computationally efficient, and helps deep networks train faster.

Deeper understanding:
Advantages:

- Faster computation
- Reduces vanishing gradient problem
- Improves deep learning performance

If input is negative:

- output becomes 0

If positive:

- output remains same

---

24. What is tanh activation function?

Answer:

Tanh (hyperbolic tangent) is an activation function whose output ranges from -1 to 1.

Equation:

$$tanh(x)=\frac{e^x-e^{-x}}{e^x+e^{-x}}$$

Deeper understanding:
Compared to sigmoid:

- tanh is zero-centered
- often performs better during training

Uses:

- Hidden layers in neural networks

---

25. What is softmax activation function?

Answer:

Softmax converts outputs into probabilities for multi-class classification problems.

Deeper understanding:
It ensures:

- all outputs lie between 0 and 1
- total probability sum equals 1

Used in:

- digit classification
- image recognition
- multi-class problems

Example:
If class probabilities are:

- cat = 0.7
- dog = 0.2
- bird = 0.1

Model predicts cat.

---

26. What is forward propagation?

Answer:

Forward propagation is the process where input data moves through the network layer by layer to generate output.

Deeper understanding:
Steps:

1. Input enters network
2. Weighted sum calculated
3. Activation function applied
4. Output produced

Purpose:

- make predictions

---

27. What is backpropagation?

Answer:

> Backpropagation is a training algorithm used to reduce error by updating weights in reverse direction.

Deeper understanding:
Steps:

1. Calculate prediction error
2. Propagate error backward
3. Update weights using gradients

Purpose:

- minimize loss function
- improve accuracy

Backpropagation is the core learning mechanism in deep learning.

---

28. Why backpropagation is important?

Answer:

> Backpropagation is important because it helps neural networks learn efficiently by minimizing prediction error.

Deeper understanding:
Without backpropagation:

- weights cannot be optimized
- network cannot learn properly

It enables:

- deep learning
- CNN training
- high accuracy models

---

29. What is gradient descent?

Answer:

> Gradient descent is an optimization algorithm used to minimize loss by updating weights gradually.

Deeper understanding:
It moves weights toward minimum error.

Update rule:

$$w=w-\eta\frac{\partial L}{\partial w}$$

Where:

- (L) = loss function
- (η) = learning rate

Purpose:

- reduce training error
- improve model performance

---

30. What is loss function?

Answer:

> Loss function measures how far predicted output is from actual output.

Deeper understanding:
Higher loss:

- poor prediction

Lower loss:

- better prediction

Examples:

- Mean Squared Error (MSE)
- Cross Entropy Loss

Training aims to minimize loss.

---

31. Difference between loss and accuracy?

Loss	Accuracy
Measures prediction error	Measures correct predictions

Lower is better	Higher is better
Used during optimization	Used for evaluation

Deeper understanding:
A model can have:

- low loss and high accuracy → good model

---

32. What is optimization in neural networks?

Answer:

> Optimization is the process of finding the best weights and bias values that minimize loss.

Deeper understanding:
Optimization algorithms:

- Gradient Descent
- Adam
- RMSProp

Purpose:

- faster convergence
- improved accuracy

---

33. What is overfitting?

Answer:

> Overfitting occurs when a model memorizes training data and performs poorly on new unseen data.

Deeper understanding:
Symptoms:

- very high training accuracy
- low testing accuracy

Causes:

- too much training

- complex model
- less data

---

34. What is underfitting?

Answer:

> Underfitting occurs when the model cannot learn patterns properly from training data.

Deeper understanding:
Symptoms:

- low training accuracy
- low testing accuracy

Causes:

- very simple model
- insufficient training
- poor feature learning

---

35. How to reduce overfitting?

Answer:

> Overfitting can be reduced using regularization techniques and better training methods.

Methods:

- Dropout
- More training data
- Early stopping
- Data augmentation
- Regularization

Deeper understanding:
Goal:

- improve generalization
- perform well on unseen data

Dropout randomly disables neurons during training to prevent memorization.

36. What is logistic regression?

Answer:

> Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

Deeper understanding:
It predicts probability between 0 and 1 using sigmoid activation.

Examples:

- spam or not spam
- disease or no disease

Sigmoid equation:

$$\sigma(x)=\frac{1}{1+e^{-x}}$$

---

37. Why sigmoid is used in logistic regression?

Answer:

> Sigmoid converts output into probability values between 0 and 1.

Deeper understanding:
If probability > 0.5:

- class = 1

Else:

- class = 0

This makes sigmoid ideal for binary classification.

---

38. Difference between linear and logistic regression?

Linear Regression	Logistic Regression
Used for prediction	Used for classification
Output is continuous	Output is probability
Uses straight line	Uses sigmoid curve

Deeper understanding:
Linear regression predicts values like salary or temperature.
Logistic regression predicts categories like yes/no.

---

39. What is binary classification?

Answer:

> Binary classification means classifying data into two classes only.

Examples:

- 0 or 1
- true or false
- spam or not spam

Deeper understanding:
Models used:

- perceptron
- logistic regression
- binary neural networks

---

40. Explain confusion matrix.

Answer:

> Confusion matrix is a table used to evaluate classification performance.

It contains:

- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Deeper understanding:
It helps calculate:

- accuracy
- precision
- recall

---

41. What is precision?

Answer:

> Precision measures how many predicted positive values are actually correct.

Formula:

$$Precision=\frac{TP}{TP+FP}$$

Deeper understanding:
High precision means fewer false positives.

---

42. What is recall?

Answer:

> Recall measures how many actual positive values are correctly identified.

Formula:

$$Recall=\frac{TP}{TP+FN}$$

Deeper understanding:
High recall means fewer false negatives.

---

43. What is accuracy?

Answer:

> Accuracy is the percentage of correctly predicted outputs.

Formula:

$$Accuracy=\frac{Correct\ Predictions}{Total\ Predictions}$$

Deeper understanding:
Higher accuracy means better model performance.

---

44. What is AUC score?

Answer:

AUC (Area Under Curve) measures how well a classifier separates classes.

Deeper understanding:

- Higher AUC → better classifier
- AUC close to 1 → excellent model
- AUC close to 0.5 → random prediction

CNN / Deep Learning

45. What is CNN?

Answer:

CNN stands for Convolutional Neural Network and is mainly used for image processing tasks.

Deeper understanding:
CNN automatically extracts image features like:

- edges
- shapes
- textures

Applications:

- face recognition
- handwritten digit recognition
- object detection

---

46. Why CNN is used for images?

Answer:

CNN is used for images because it can automatically detect important visual features.

Deeper understanding:
Unlike traditional ANN:

- CNN preserves spatial information
- requires fewer parameters
- performs better on image data

---

47. What is convolution?

Answer:

> Convolution is the process of applying filters to extract important features from images.

Deeper understanding:
Filters scan the image and detect:

- edges
- corners
- patterns

This creates feature maps.

---

48. What is filter/kernel?

Answer:

> Filter or kernel is a small matrix used in convolution to detect image features.

Deeper understanding:
Different filters detect:

- vertical edges
- horizontal edges
- textures

Example:
3×3 filter matrix.

---

49. What is feature map?

Answer:

> Feature map is the output produced after applying convolution on an image.

Deeper understanding:
Feature maps contain detected patterns and important image information.

---

50. What is pooling?

Answer:

> Pooling reduces feature map size and decreases computation.

Deeper understanding:
Benefits:

- faster computation
- reduced overfitting
- memory efficiency

Types:

- max pooling
- average pooling

---

51. Difference between max pooling and average pooling?

Max Pooling	Average Pooling
Takes maximum value	Takes average value
Preserves strongest features	Smooths features
Mostly used	Less commonly used

Deeper understanding:
Max pooling performs better in CNNs.

---

52. What is flatten layer?

Answer:

> Flatten layer converts 2D feature maps into 1D vectors before fully connected layers.

Deeper understanding:
CNN layers output matrices.
Dense layers require vectors, so flattening is necessary.

---

53. What is fully connected layer?

Answer:

> Fully connected layer connects every neuron with all neurons in the next layer.

Deeper understanding:
Purpose:

- final classification
- decision making

Usually present at the end of CNN.

---

54. Why CNN performs better on images?

Answer:

> CNN performs better because it automatically learns spatial and visual features from images.

Deeper understanding:
Advantages:

- fewer parameters
- feature extraction
- translation invariance
- high accuracy

---

55. What is MNIST dataset?

Answer:

> MNIST is a dataset of handwritten digit images from 0 to 9.

Deeper understanding:

- 28×28 grayscale images
- 10 classes
- widely used for CNN practice

---

56. How many classes are in MNIST?

Answer:

MNIST contains 10 classes representing digits 0 to 9.

---

57. Why MNIST is popular?

Answer:

MNIST is popular because it is simple, clean, and ideal for learning deep learning concepts.

Deeper understanding:
Used for:

- ANN practice
- CNN implementation
- benchmarking models

---

58. Input size of MNIST images?

Answer:

MNIST images are 28 × 28 grayscale images.

Total pixels:

- 784 input values

---

59. What preprocessing did you perform?

Answer:

Data preprocessing included normalization and reshaping.

Deeper understanding:
Normalization scales pixel values between 0 and 1 for faster training.

TensorFlow / PyTorch

60. What is TensorFlow?

Answer:

> TensorFlow is an open-source deep learning framework developed by Google.

Uses:

- neural networks
- deep learning
- AI applications

---

61. What is Keras?

Answer:

> Keras is a high-level API built on TensorFlow used for easy neural network implementation.

Deeper understanding:
Advantages:

- simple syntax
- less code
- beginner friendly

---

62. Difference between TensorFlow and Keras?

TensorFlow	Keras
Complete framework	High-level API
More complex	Easier to use
Flexible	Beginner friendly

Deeper understanding:
Keras runs on top of TensorFlow.

---

63. Why TensorFlow warning about GPU appears?

Answer:

> The warning appears because TensorFlow GPU support is not available on native Windows for newer versions.

Deeper understanding:
TensorFlow still works using CPU, so it is not dangerous.

---

64. CPU vs GPU in deep learning?

CPU	GPU
Fewer cores	Thousands of cores
Slower training	Faster training
Good for small tasks	Good for deep learning

Deeper understanding:
GPU performs parallel computation efficiently.

---

65. What is tensor?

Answer:

> Tensor is a multi-dimensional data structure used in deep learning.

Examples:

- scalar → 0D tensor
- vector → 1D tensor
- matrix → 2D tensor

---

66. What is PyTorch?

Answer:

> PyTorch is an open-source deep learning framework developed by Meta.

Deeper understanding:
Popular for:

- research
- experimentation
- dynamic computation graphs

---

67. Difference between TensorFlow and PyTorch?

TensorFlow	PyTorch
Production focused	Research focused
Static graph	Dynamic graph
More deployment tools	Easier debugging

---

68. Which framework is easier?

Answer:

> PyTorch is generally considered easier for beginners and research, while Keras is easiest for quick implementation.

---

69. Why PyTorch is popular in research?

Answer:

> PyTorch is popular because it supports dynamic graphs and easier debugging.

Deeper understanding:
Researchers prefer PyTorch because:

- code is flexible
- experiments are easier
- debugging is simpler

70. What is XOR problem in perceptron?

Answer:

> XOR problem is a classification problem that cannot be solved by a single-layer perceptron because the data is not linearly separable.

XOR truth table:

Input	Output
0 0	0
0 1	1
1 0	1
1 1	0

Deeper understanding:
No single straight line can separate XOR outputs correctly.

Solution:

- Multi-Layer Perceptron (MLP)
- Hidden layers
- Non-linear activation functions

---

71. Why multi-layer perceptron solves XOR?

Answer:

> Multi-layer perceptron solves XOR because hidden layers and non-linear activation functions allow non-linear decision boundaries.

Deeper understanding:
Hidden layers learn complex feature combinations, making non-linear classification possible.

---

72. What is hidden layer?

Answer:

> Hidden layer is the layer between input and output layers where feature learning occurs.

Deeper understanding:

- Input layer receives data
- Hidden layers process information
- Output layer gives prediction

Deep networks contain multiple hidden layers.

---

73. What is vanishing gradient problem?

Answer:

> Vanishing gradient problem occurs when gradients become extremely small during backpropagation.

Deeper understanding:
This slows or stops learning in deep neural networks.

Mostly happens with:

- sigmoid
- tanh activation

Solution:

- ReLU activation
- batch normalization

---

74. What is dropout?

Answer:

> Dropout is a regularization technique where some neurons are randomly disabled during training.

Deeper understanding:
Purpose:

- reduce overfitting
- improve generalization

Example:
During training, some neurons are ignored temporarily so the model does not memorize data.

---

75. What is batch normalization?

Answer:

> Batch normalization normalizes layer inputs during training to improve speed and stability.

Deeper understanding:
Advantages:

- faster convergence
- stable training
- reduced vanishing gradient

---

76. Difference between supervised and unsupervised learning?

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Predicts output	Finds hidden patterns
Example: classification	Example: clustering

Deeper understanding:
Supervised learning learns from known answers, while unsupervised learning discovers structure automatically.

---

77. What is reinforcement learning?

Answer:

> Reinforcement learning is a learning method where an agent learns by interacting with environment using rewards and penalties.

Deeper understanding:
The model improves by maximizing reward.

Applications:

- robotics
- game playing
- self-driving cars

---

78. Difference between AI, ML, and DL?

AI	ML	DL
Broad field	Subset of AI	Subset of ML
Simulates intelligence	Learns from data	Uses deep neural networks

Deeper understanding:

- AI → overall intelligence
- ML → learning algorithms
- DL → neural network-based learning

---

79. What is hyperparameter?

Answer:

> Hyperparameters are settings defined before training that control the learning process.

Examples:

- learning rate
- epochs
- batch size
- number of layers

Deeper understanding:
Hyperparameters are not learned automatically.

---

80. Explain bias-variance tradeoff.

Answer:

> Bias-variance tradeoff balances underfitting and overfitting.

Deeper understanding:

- High bias → underfitting
- High variance → overfitting

Goal:

- achieve good generalization

Code-Based Questions

81. Why `np.zeros(2)` used?

Answer:

> `np.zeros(2)` initializes two weights with value 0.

Deeper understanding:
Since the perceptron has two input features, two initial weights are needed.

---

82. Why `epochs=50` chosen?

Answer:

> `epochs=50` gives enough iterations for the model to learn properly.

Deeper understanding:
Training may stop earlier if convergence occurs before 50 epochs.

---

83. Why use `plt.contourf()`?

Answer:

> `plt.contourf()` is used to visualize decision regions graphically.

Deeper understanding:
It fills different classification regions with different colors.

---

84. Why reshape is required?

Answer:

> Reshaping converts data into required dimensions for neural network input.

Deeper understanding:
CNN models usually expect:

- height
- width
- channels

So reshaping becomes necessary.

---

85. What does `model.fit()` do?

Answer:

> `model.fit()` trains the neural network using training data.

Deeper understanding:
It performs:

- forward propagation
- loss calculation
- backpropagation
- weight updates

---

86. Why use `predict()` separately?

Answer:

> `predict()` is used to generate outputs for new or test data after training.

Deeper understanding:
Training and prediction are separate processes.

---

87. What is `np.dot()`?

Answer:

`np.dot()` performs dot product operation between vectors or matrices.

Equation:

z=w\cdot x+b

Deeper understanding:
It calculates weighted sum inside neurons.

---

88. Why normalize data?

Answer:

> Normalization scales data into smaller ranges for faster and stable training.

Deeper understanding:
Benefits:

- faster convergence
- reduced training instability
- improved performance

Example:
Pixel values scaled from:
0–255 → 0–1

---

89. Why train-test split is needed?

Answer:

> Train-test split evaluates model performance on unseen data.

Deeper understanding:

- training set → learning
- testing set → evaluation

Purpose:

- detect overfitting
- measure generalization

---

90. Why use ReLU in hidden layer?

Answer:

> ReLU is used because it is computationally efficient and reduces vanishing gradient problem.

ReLU equation:

$f(x)=\max(0,x)$

Deeper understanding:
Advantages:

- faster training
- simple computation
- better deep learning performance

Most Important Final Viva Questions (High Probability)

These are the questions externals ask most frequently in ANN practical exams.

1. Explain Perceptron Learning Law.

Answer:

> Perceptron learning law updates weights and bias whenever prediction is incorrect to reduce classification error.

Learning rule:

$w_{new}=w_{old}+\eta(y-\hat{y})x$

Key points:

- If prediction correct → no update
- If prediction wrong → weights adjusted
- Helps model learn decision boundary

2. Why perceptron cannot solve XOR problem?

Answer:

> XOR is not linearly separable, and perceptron can only create linear decision boundaries.

Important line for viva:

> “No single straight line can separate XOR outputs.”

Solution:

- Multi-layer perceptron
- Hidden layers
- Non-linear activation functions

3. What is backpropagation?

Answer:

> Backpropagation is the process of updating neural network weights by propagating error backward to minimize loss.

Steps:

1. Forward propagation
2. Calculate error
3. Backward propagation
4. Update weights

Importance:

- Core learning algorithm in deep learning

4. Difference between ANN and CNN?

ANN	CNN
General-purpose neural network	Specialized for image processing
Fully connected layers	Uses convolution and pooling
More parameters	Fewer parameters
Less efficient for images	Better for image recognition

Most important line:

> “CNN automatically extracts image features.”

5. What is activation function and why is it needed?

Answer:

> Activation function introduces non-linearity so neural networks can solve complex problems.

Without activation:

- network behaves like linear model
- cannot solve complex tasks

Most common:
ReLU

f(x)=\max(0,x)

6. What is overfitting?

Answer:

> Overfitting occurs when the model memorizes training data and performs poorly on unseen data.

Symptoms:

- high training accuracy
- low testing accuracy

Solutions:

- dropout
- more data
- regularization

7. What is CNN?

Answer:

> CNN is a deep learning architecture mainly used for image processing tasks.

Main components:

- convolution layer
- pooling layer
- fully connected layer

Applications:

- face recognition
- handwritten digit recognition
- object detection

8. What is TensorFlow and PyTorch?

TensorFlow

> TensorFlow is a deep learning framework developed by Google.

PyTorch

> PyTorch is a deep learning framework developed by Meta.

Difference:

- TensorFlow → production
- PyTorch → research and debugging

9. What is learning rate?

Answer:

> Learning rate controls how much weights are updated during training.

Symbol:

\eta

Important:

- Too high → unstable learning
- Too low → slow learning

10. Explain your practical in simple words.

Very important for external.

Perceptron Practical

> "In this practical, I implemented perceptron learning algorithm for binary classification. The model learned by updating weights using perceptron learning law. I also plotted decision regions and decision boundary graphically."

CNN/MNIST Practical

> "In this practical, I trained CNN models using Keras, TensorFlow, and PyTorch on the MNIST handwritten digit dataset and compared their accuracies."

AUltimate Emergency Viva Lines

If nervous, remember these safe lines:

About ANN

> “ANN learns patterns from data by adjusting weights and bias.”

About CNN

> “CNN is mainly used for image processing.”

About Backpropagation

> “Backpropagation minimizes error by updating weights.”

About Epoch

> “One complete pass of training data is called an epoch.”

About Activation Function

> “Activation function introduces non-linearity.”